

Lab 1: Packet analysis at application layer using Wireshark
SCSR1213 Network Communications
Universiti Teknologi Malaysia

Objective:

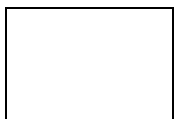
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce students with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Metric No : A23CS0220

Section : 02



Mark

PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

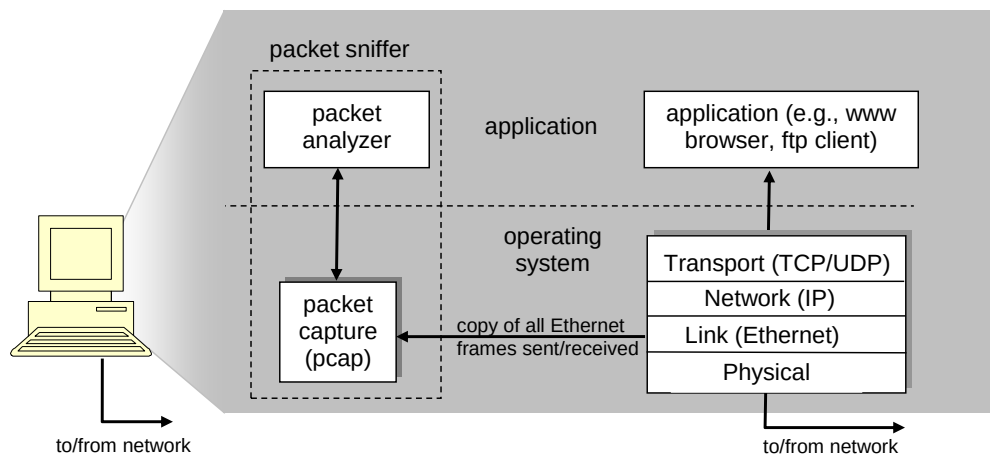


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

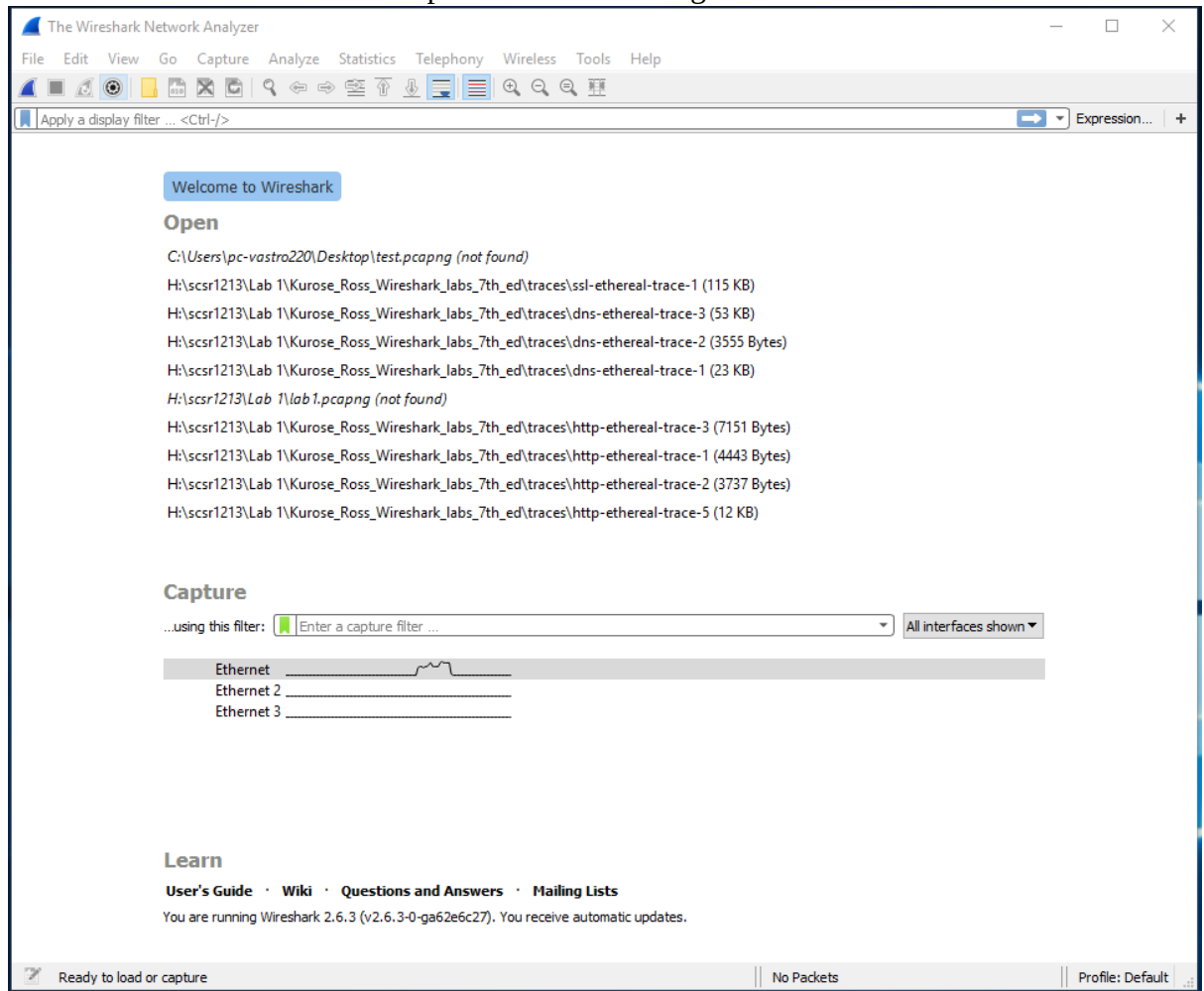


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

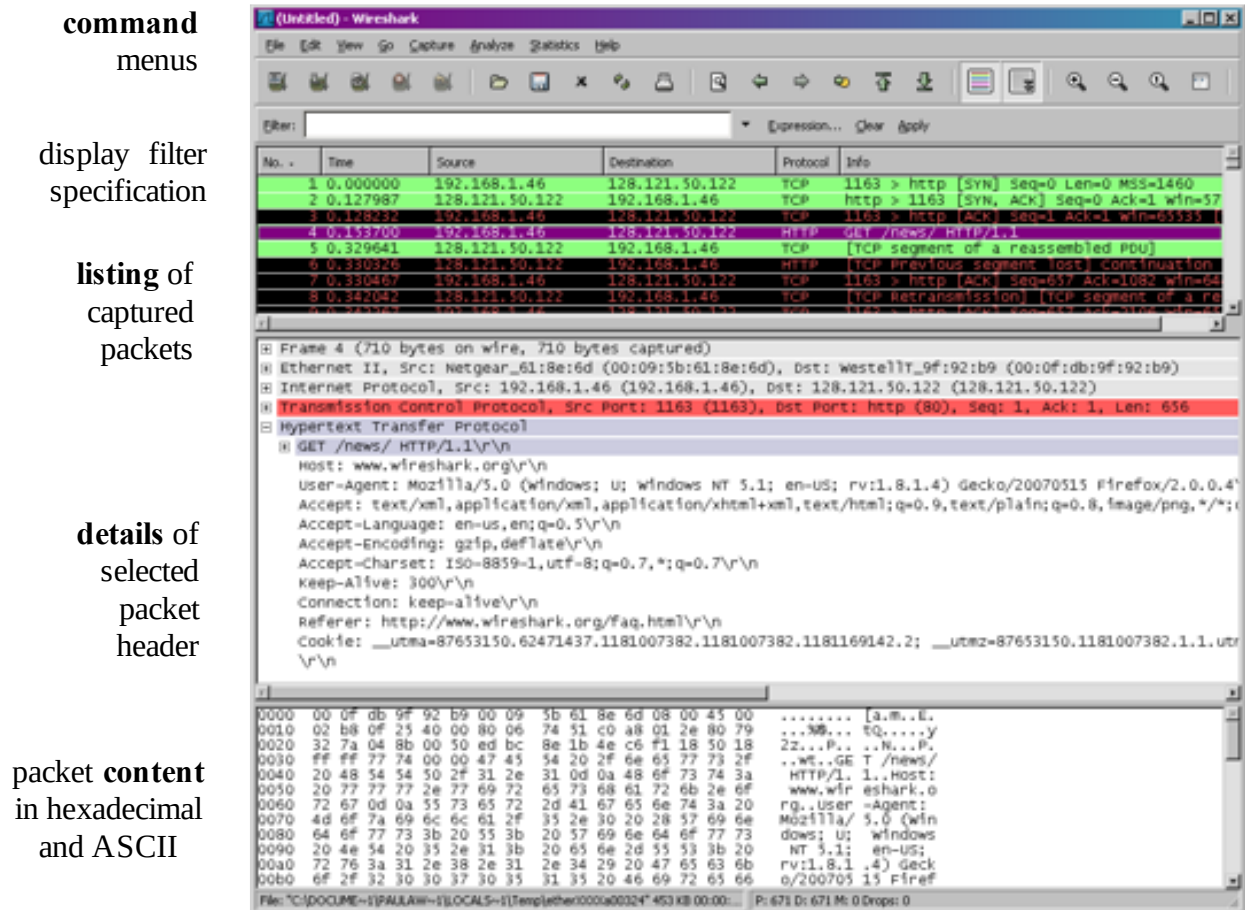


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

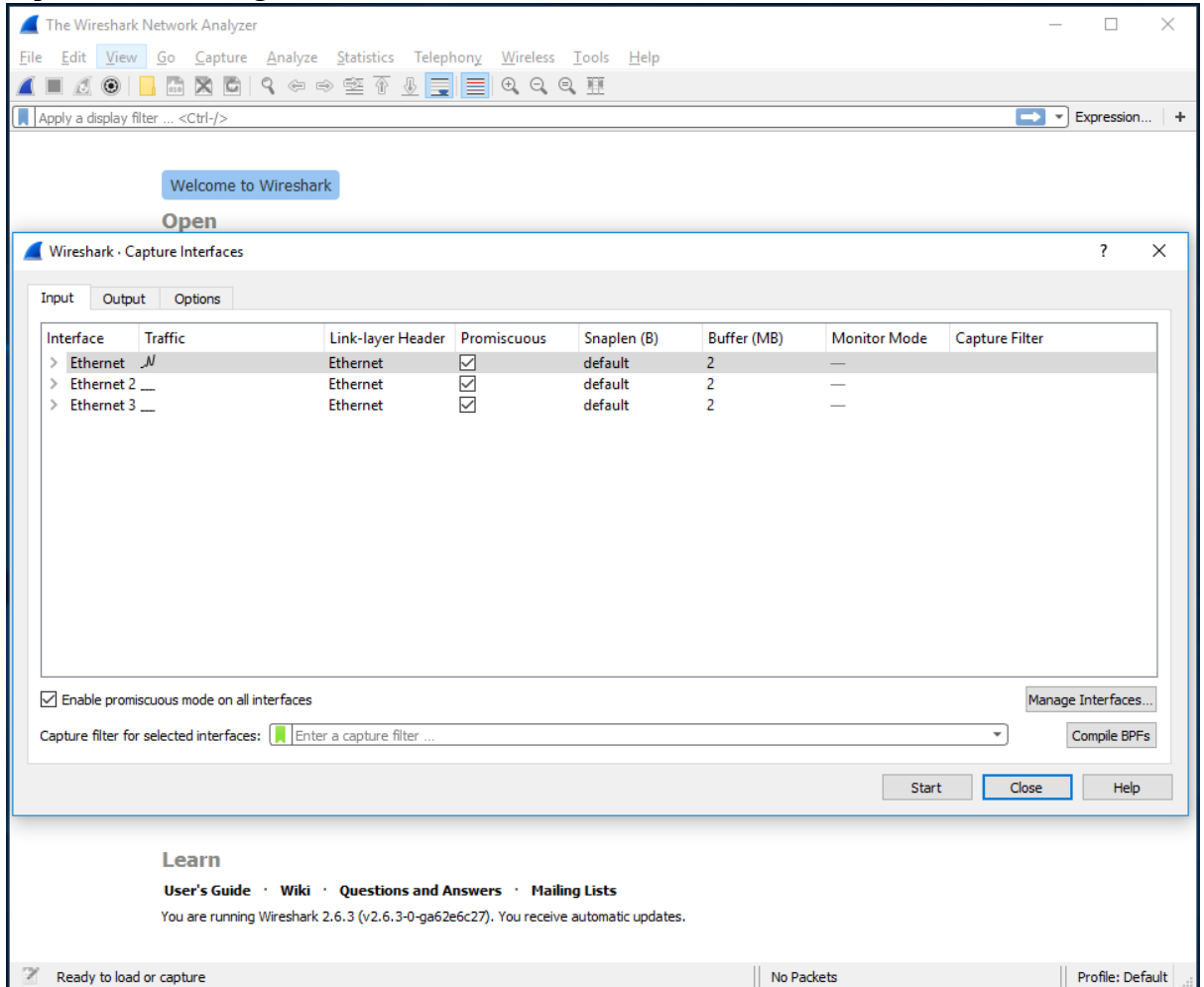
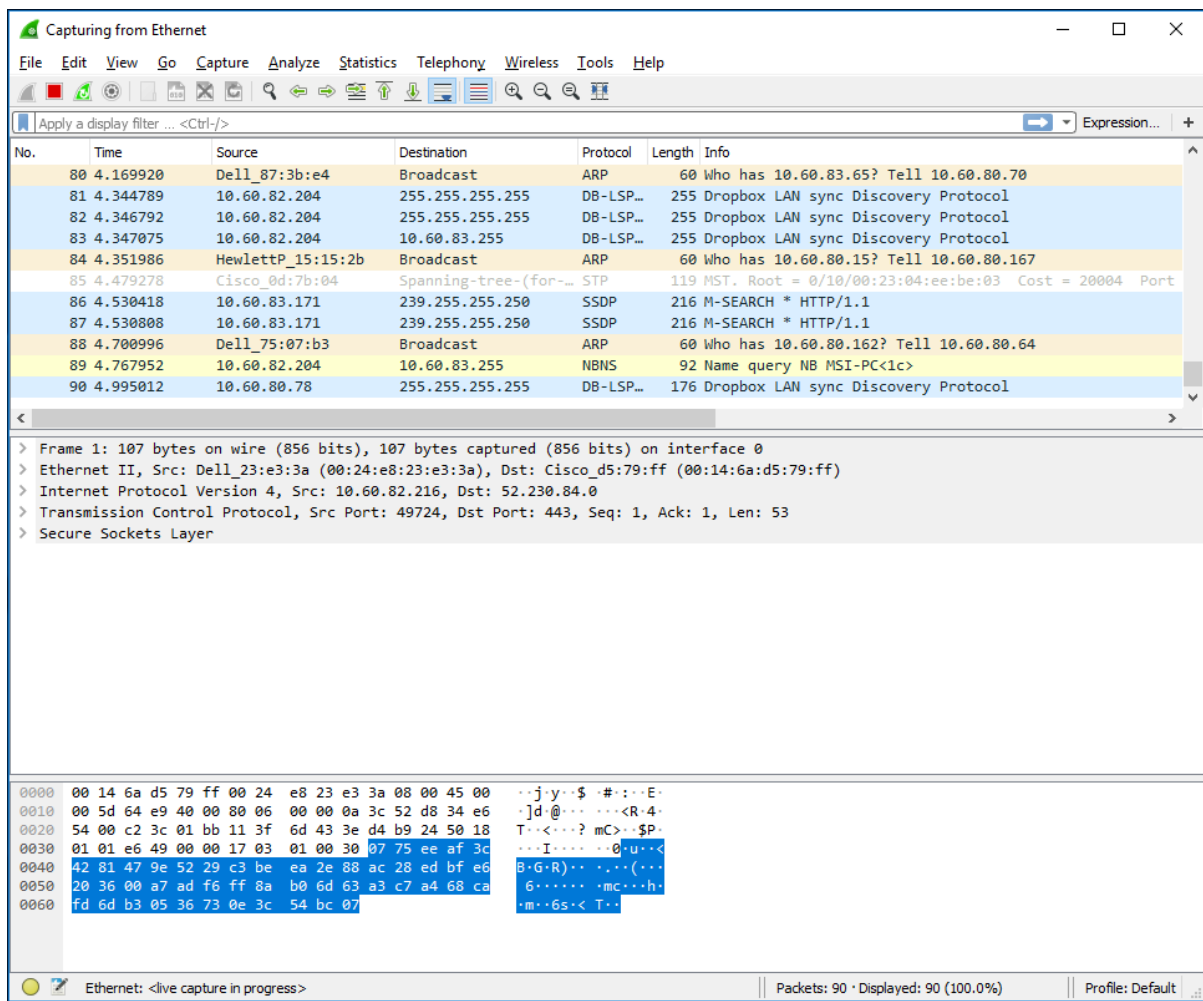


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.



The image shows the Wireshark interface with a packet capture in progress. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. A display filter bar shows 'Apply a display filter ... <Ctrl-/>'. The main packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
80	4.169920	Dell_87:3b:e4	Broadcast	ARP	60	Who has 10.60.83.65? Tell 10.60.80.70
81	4.344789	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
82	4.346792	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
83	4.347075	10.60.82.204	10.60.83.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
84	4.351986	HewlettP_15:15:2b	Broadcast	ARP	60	Who has 10.60.80.15? Tell 10.60.80.167
85	4.479278	Cisco_0d:7b:04	Spanning-tree-(for-...	STP	119	MST. Root = 0/10/00:23:04:ee:be:03 Cost = 20004 Port
86	4.530418	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
87	4.530808	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
88	4.700996	Dell_75:07:b3	Broadcast	ARP	60	Who has 10.60.80.162? Tell 10.60.80.64
89	4.767952	10.60.82.204	10.60.83.255	NBNS	92	Name query NB MSI-PC<1c>
90	4.995012	10.60.80.78	255.255.255.255	DB-LSP...	176	Dropbox LAN sync Discovery Protocol

Below the packet list, the details pane shows the selected packet (Frame 1) with the following information:

- > Frame 1: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface 0
- > Ethernet II, Src: Dell_23:e3:3a (00:24:e8:23:e3:3a), Dst: Cisco_d5:79:ff (00:14:6a:d5:79:ff)
- > Internet Protocol Version 4, Src: 10.60.82.216, Dst: 52.230.84.0
- > Transmission Control Protocol, Src Port: 49724, Dst Port: 443, Seq: 1, Ack: 1, Len: 53
- > Secure Sockets Layer

The bottom pane shows the raw packet data in hexadecimal and ASCII format.

Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

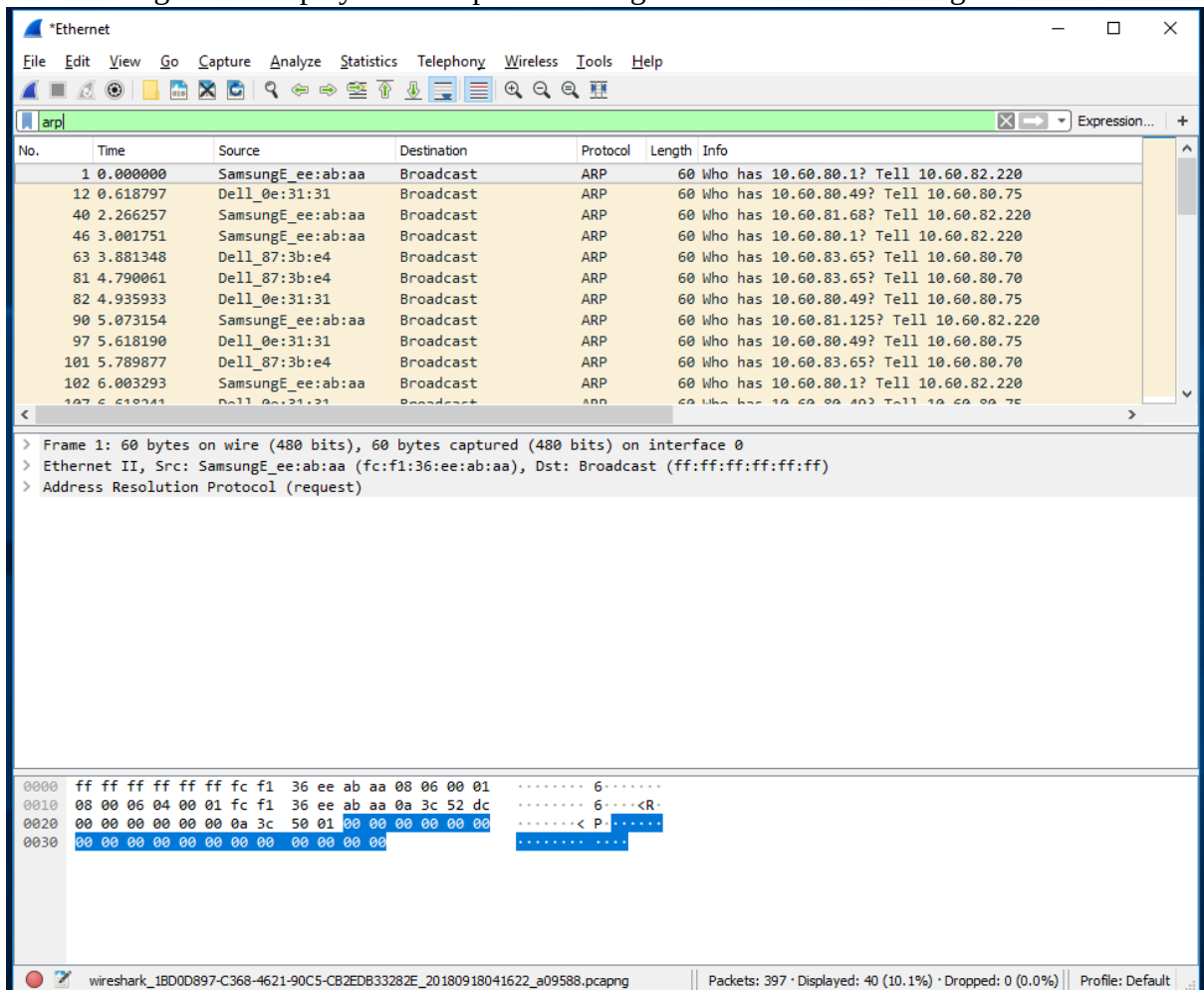


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

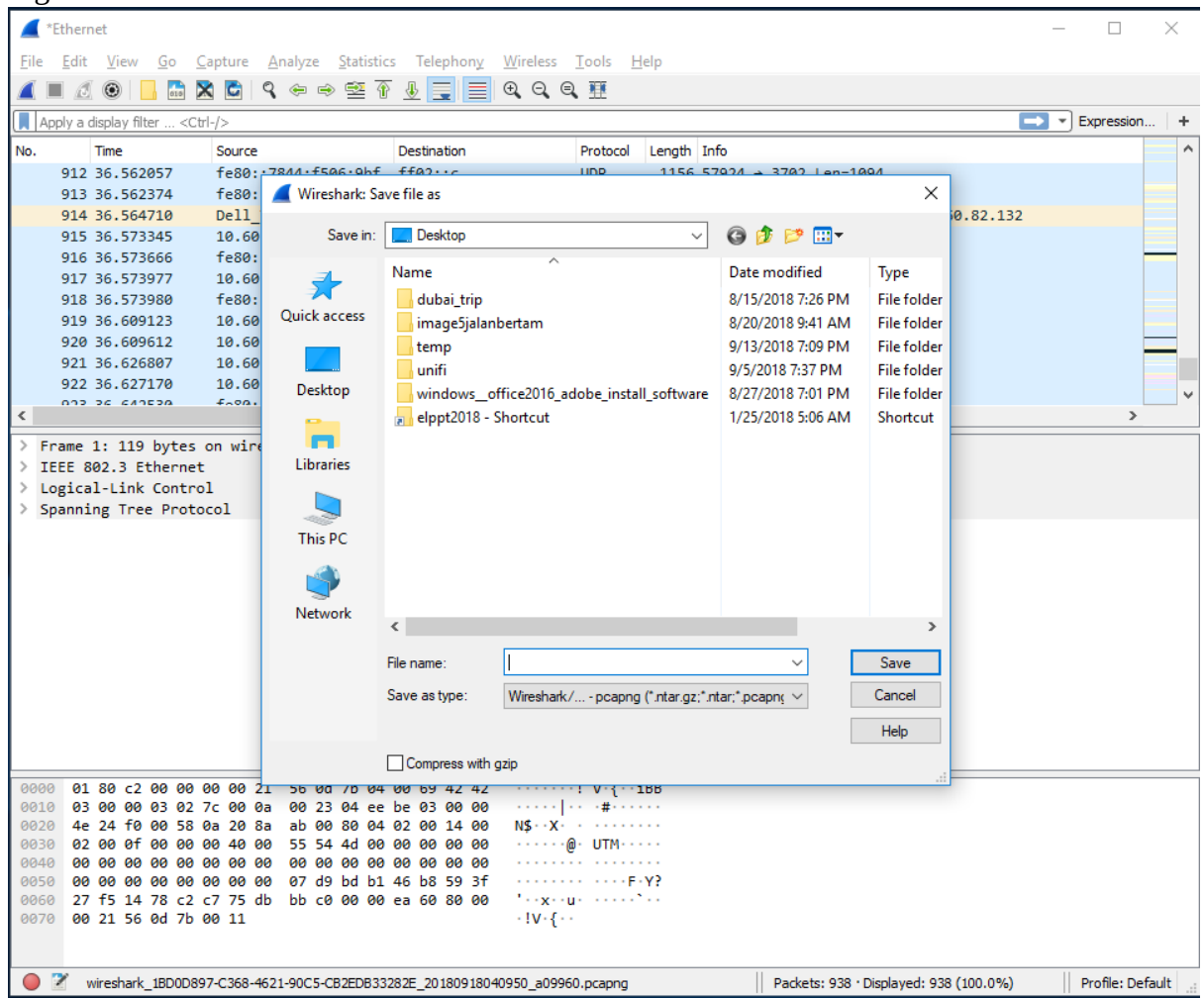


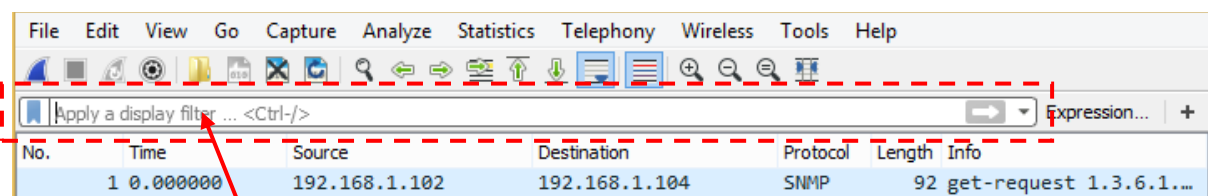
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:
 1. What version of HTTP is the server running?
HTTP/1.1
 2. What is the IP address of the client computer?
192.168.1.102
 3. What is the IP address of the gaia.cs.umass.edu server?
128.119.245.12
 4. How many bytes of content are being returned to client browser?
1071 bytes
 5. What is the status code returned from the server to client browser?
 - For packet 12, the status code is 200 OK (successful response).
 - For packet 14, the status code is 404 Not Found (indicating the resource was not found).

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:
 1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?
No. it not showing the sentences given
 2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?
After the client sent the first HTTP GET request, the server explicitly returned the file contents. This is because the server responded with a 200 OK status code, which means the request was successful and the file was delivered. Additionally, the Content-Length field in the response specifies the size of the returned file, confirming that the content was sent back to the client.
 3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?
No. It also did not show the sentences given
 4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.
In packet 10, the status code is 200 OK (a successful response). Yes , when the status code is 200 OK, the server explicitly return the contents of the file because the server successfully received and processed your request. It tells the client (like a browser) that the server is sending back the content you asked for, such as a file or a webpage, without any issues.

B.3 HTML Documents with Embedded Objects

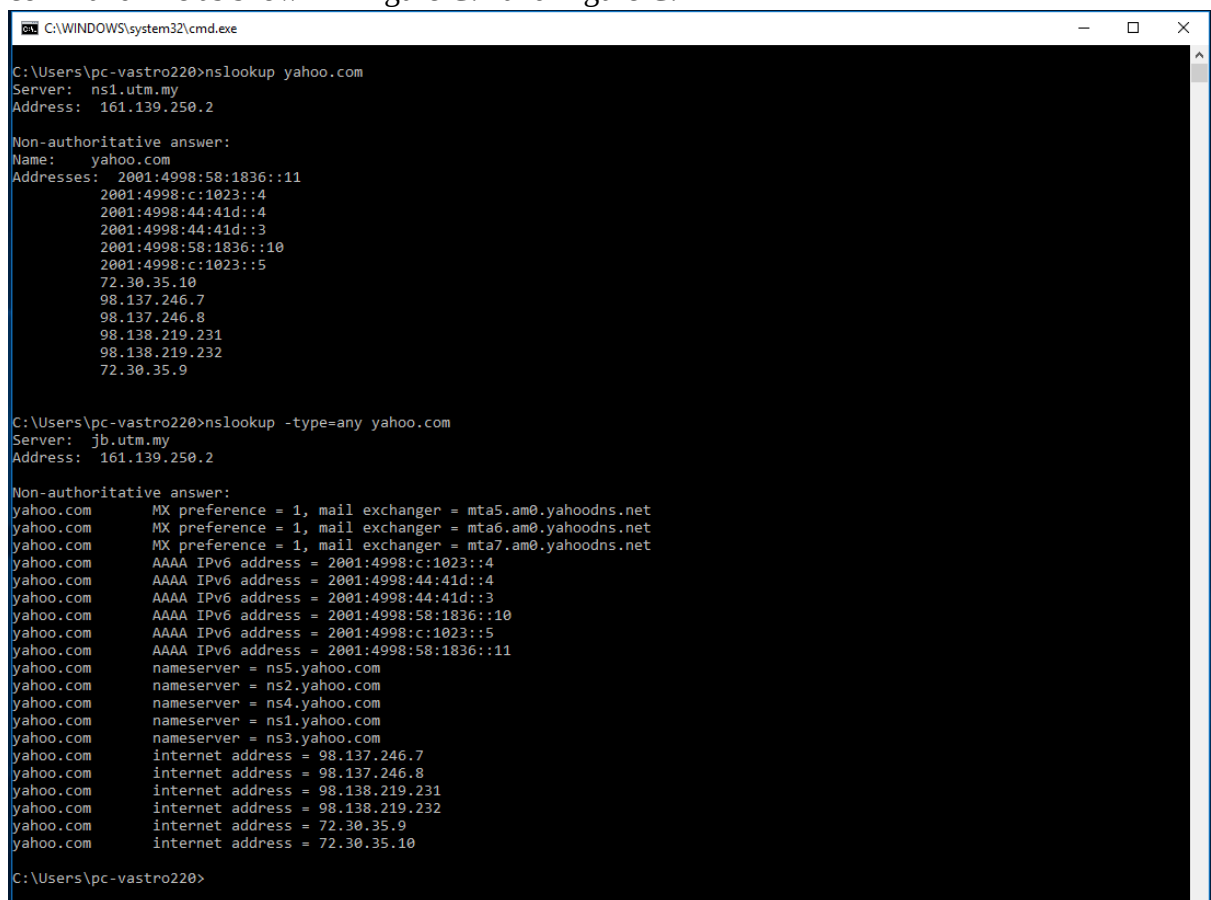
- Open packet trace file **lab1-http-B03.pcapng**.
 - By looking at the information in the HTTP GET and response messages, answer the following questions:
1. How many HTTP GET request messages did client browser send?
3 HTTP GET message
 2. To which Internet addresses were these GET requests sent?
128.119.245.12, 165.193.123.218 and 134.241.6.82
 3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?
Yes. 571 bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file
 4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?
555 bytes of content are being returned to client browser for the **cover.jpg** image file

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server:  ns1.utm.my
Address:  161.139.250.2

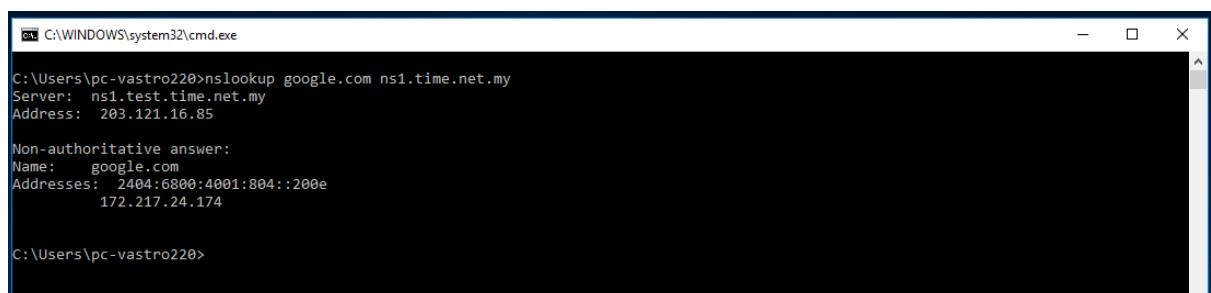
Non-authoritative answer:
Name:     yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server:  jb.utm.my
Address:  161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server:  ns1.test.time.net.my
Address:  203.121.16.85

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\L E N O V O>nslookup www.microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
Name:     e13678.dscb.akamaiedge.net
Addresses: 2001:e68:2:180::356e
           2001:e68:2:184::356e
           23.220.242.103
Aliases:  www.microsoft.com
           www.microsoft.com-c-3.edgekey.net
           www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add a screenshot to your answer.

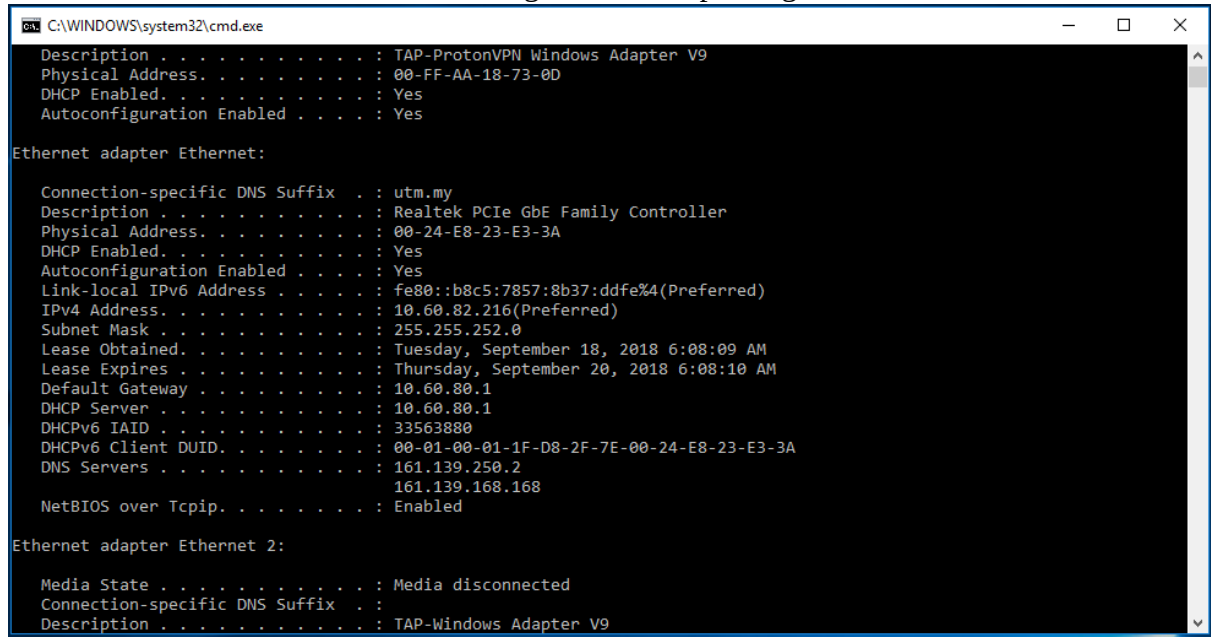
```
C:\Users\L E N O V O>nslookup microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
Name:     microsoft.com
Addresses: 2603:1030:c02:8::14
           2603:1020:201:10::10f
           2603:1010:3:3::5b
           2603:1030:b:3::152
           2603:1030:20e:3::23c
           20.112.250.133
           20.231.239.246
           20.76.201.171
           20.70.246.20
           20.236.44.162
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes

Ethernet adapter Ethernet:

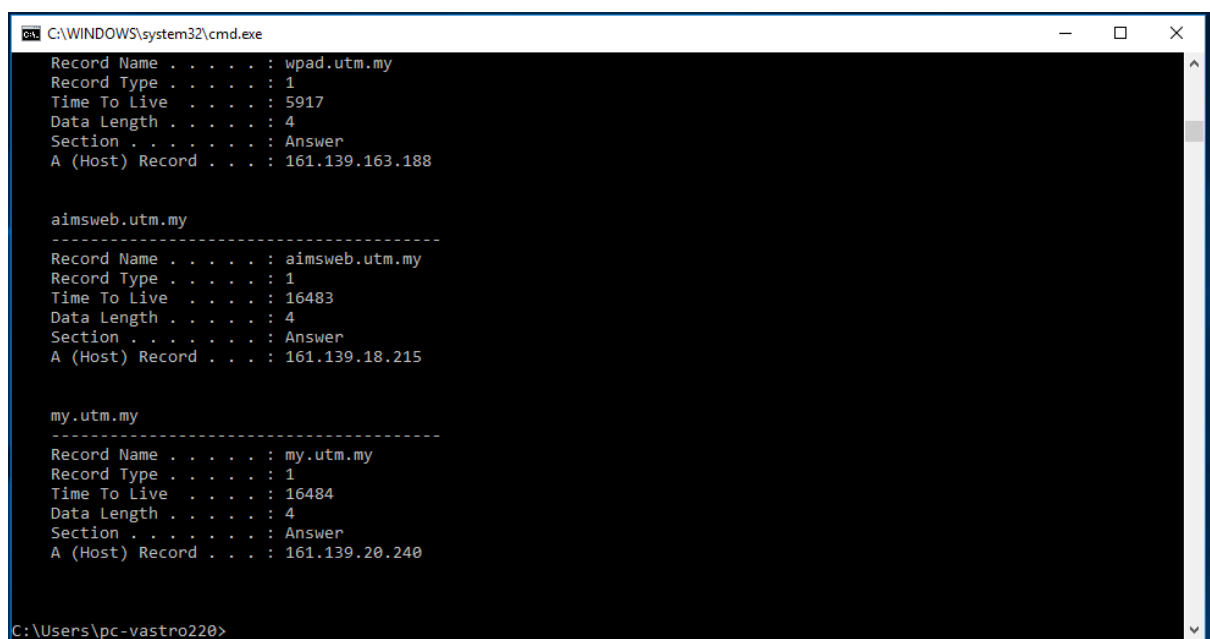
    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           : 161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds. Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

aimsweb.utm.my
-----
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

my.utm.my
-----
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

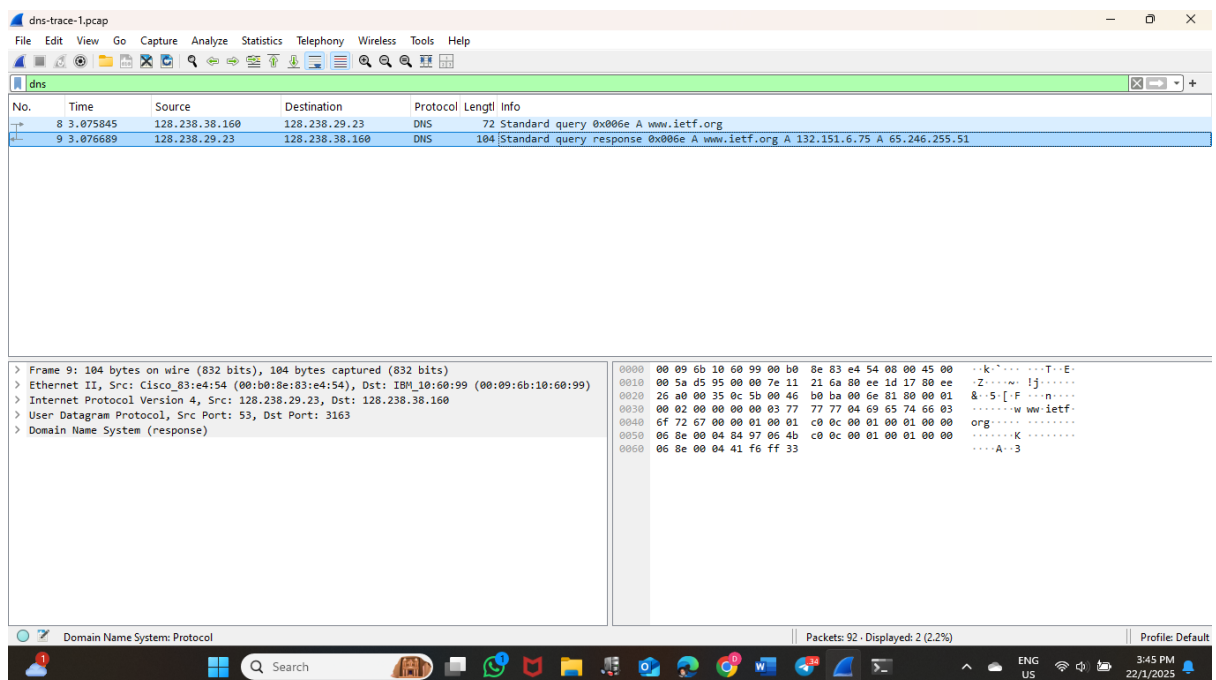
Successfully flushed the DNS Resolver Cache.

C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

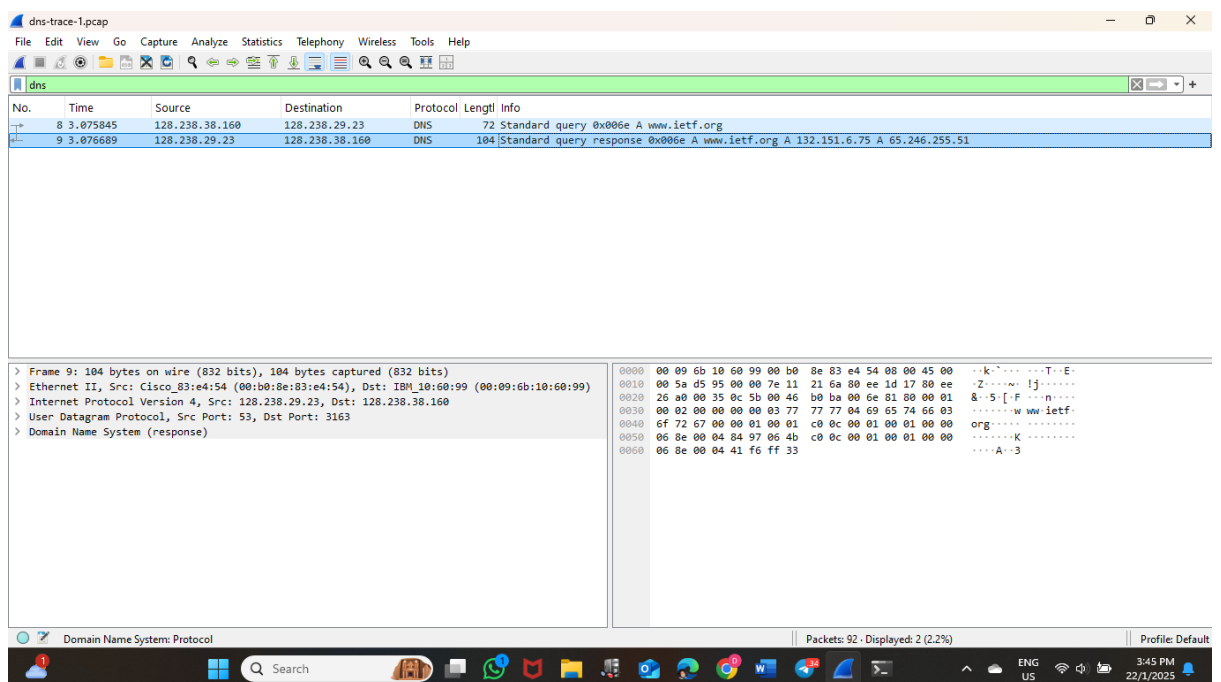
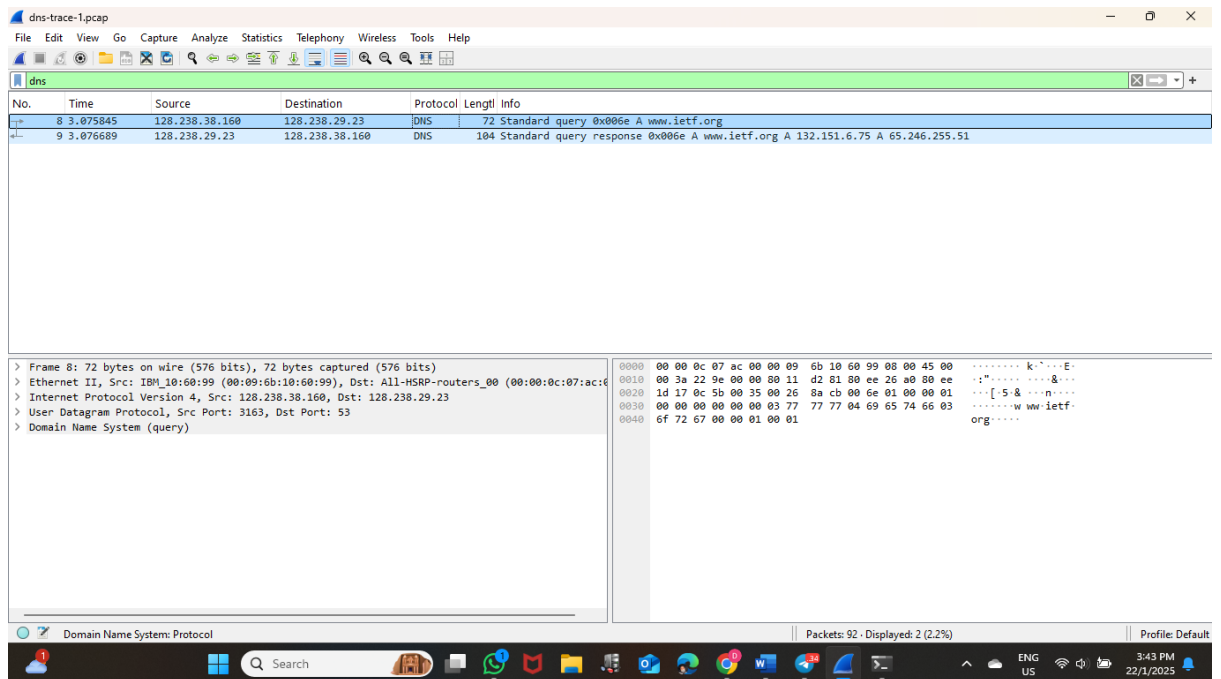
3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

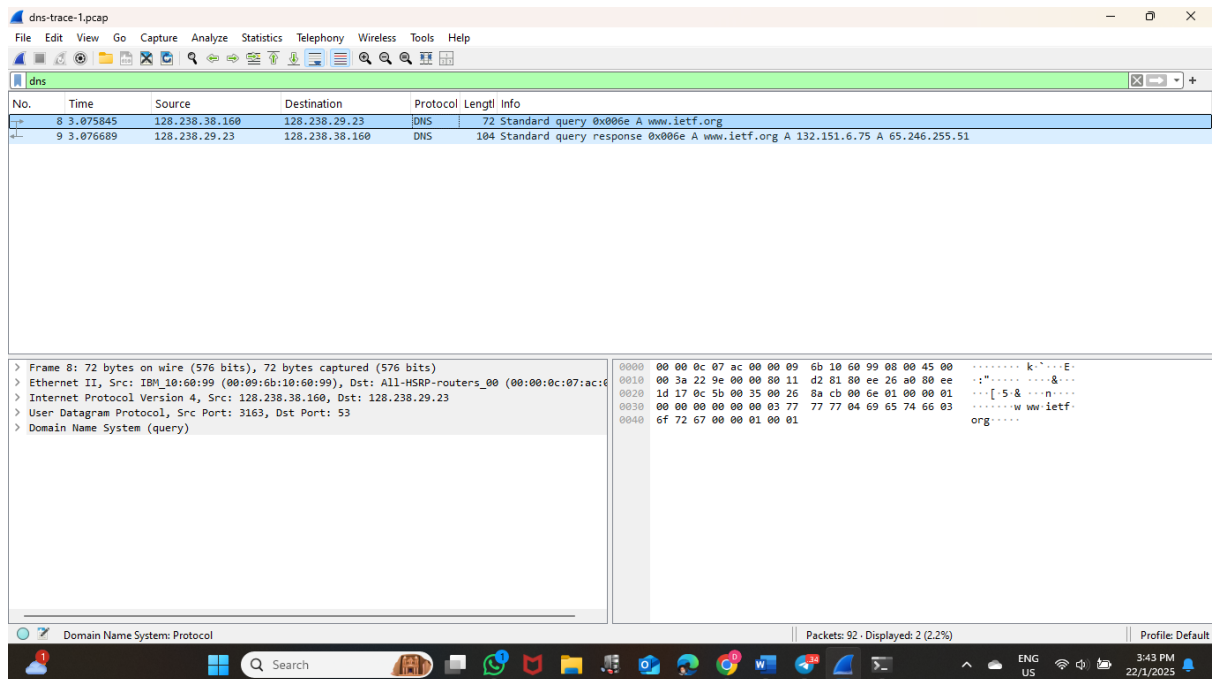


2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

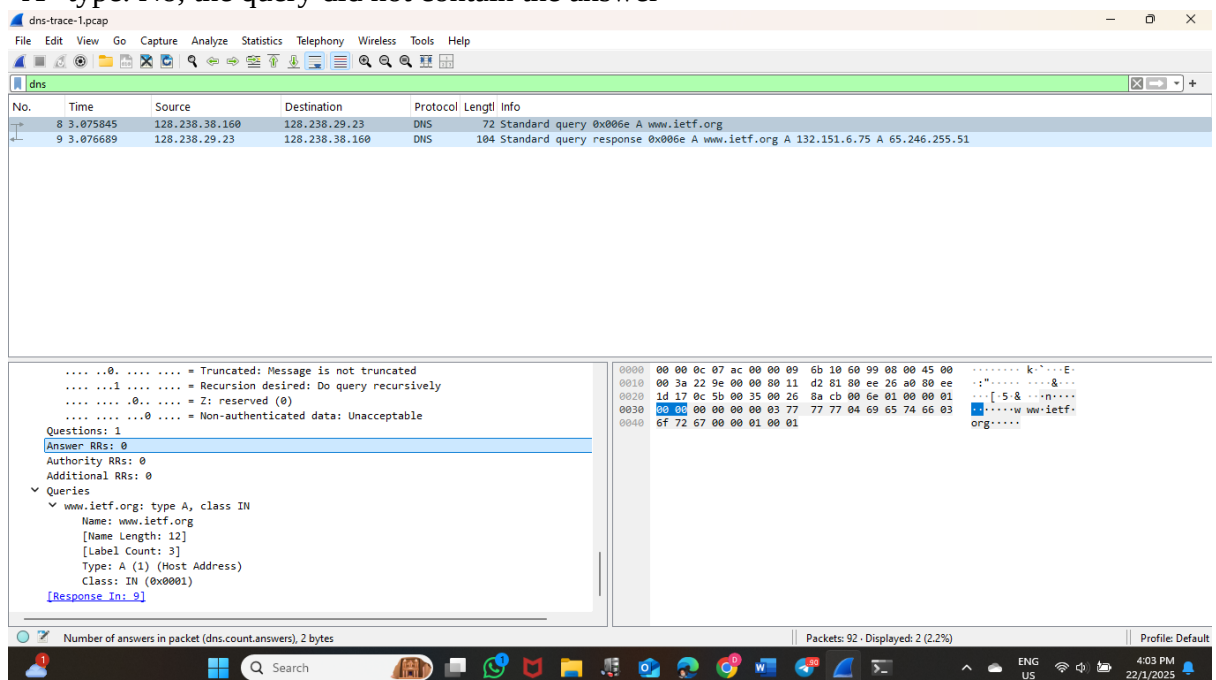
Destination port in DNS query message: 53, Source port in DNS response message: 53



3. To what IP address is the DNS query message sent? Add screenshots in your answer.
128.238.29.23



4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.
“A” type. No, the query did not contain the answer



5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.
2 answers.

The screenshot displays a Wireshark capture of a DNS response packet. The packet list at the top shows two packets: a standard query (No. 8) and a standard query response (No. 9). The details pane for packet 9 shows the DNS response structure, including the query ID, flags, and two answers. The first answer is for www.ietf.org type A, class IN, with address 132.151.6.75. The second answer is for www.ietf.org type A, class IN, with address 65.246.255.51. The packet bytes pane shows the raw data of the DNS response.

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

Class: IN (0x0001)

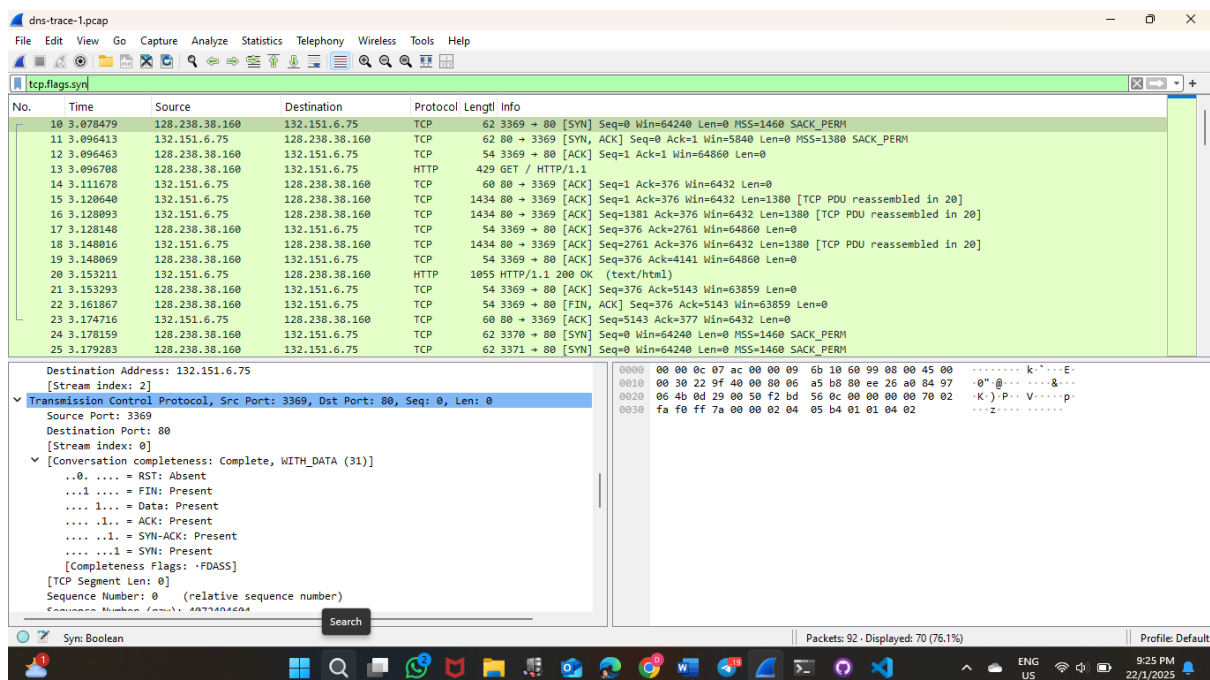
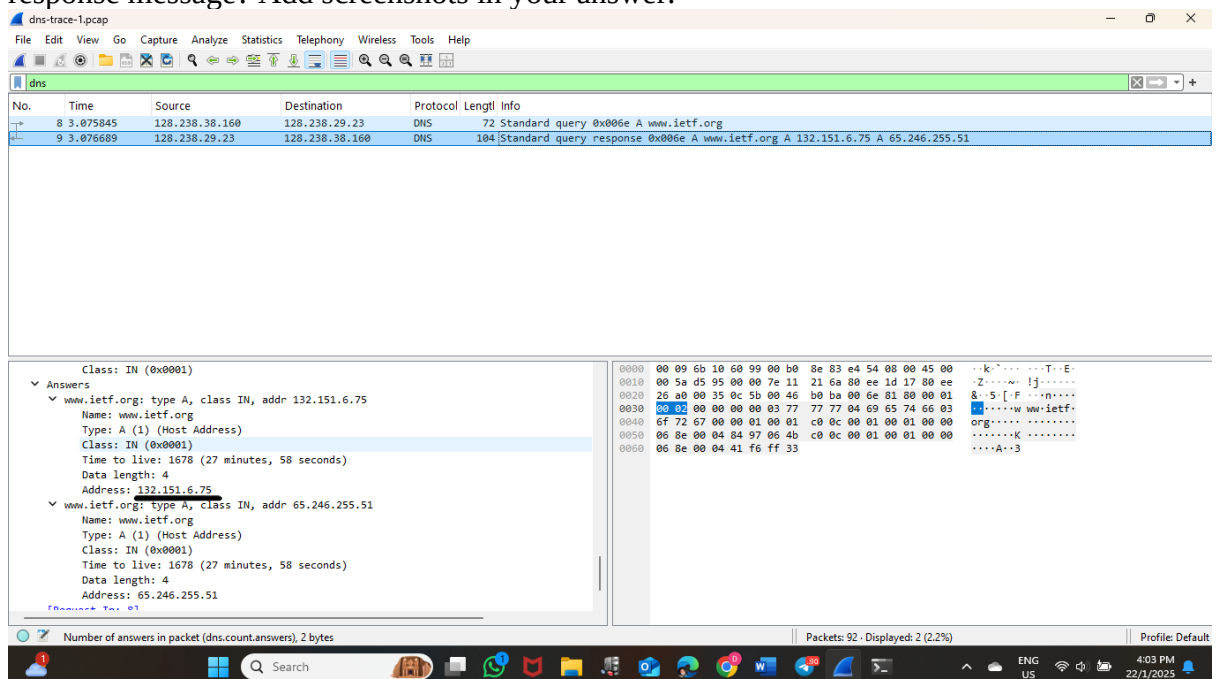
- Answers
 - www.ietf.org: type A, class IN, addr 132.151.6.75
 - Name: www.ietf.org
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
 - Time to live: 1678 (27 minutes, 58 seconds)
 - Data length: 4
 - Address: 132.151.6.75
 - www.ietf.org: type A, class IN, addr 65.246.255.51
 - Name: www.ietf.org
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
 - Time to live: 1678 (27 minutes, 58 seconds)
 - Data length: 4
 - Address: 65.246.255.51

Number of answers in packet (dns.count.answers): 2 bytes

Packets: 92 - Displayed: 2 (2.2%)

Profile: Default

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

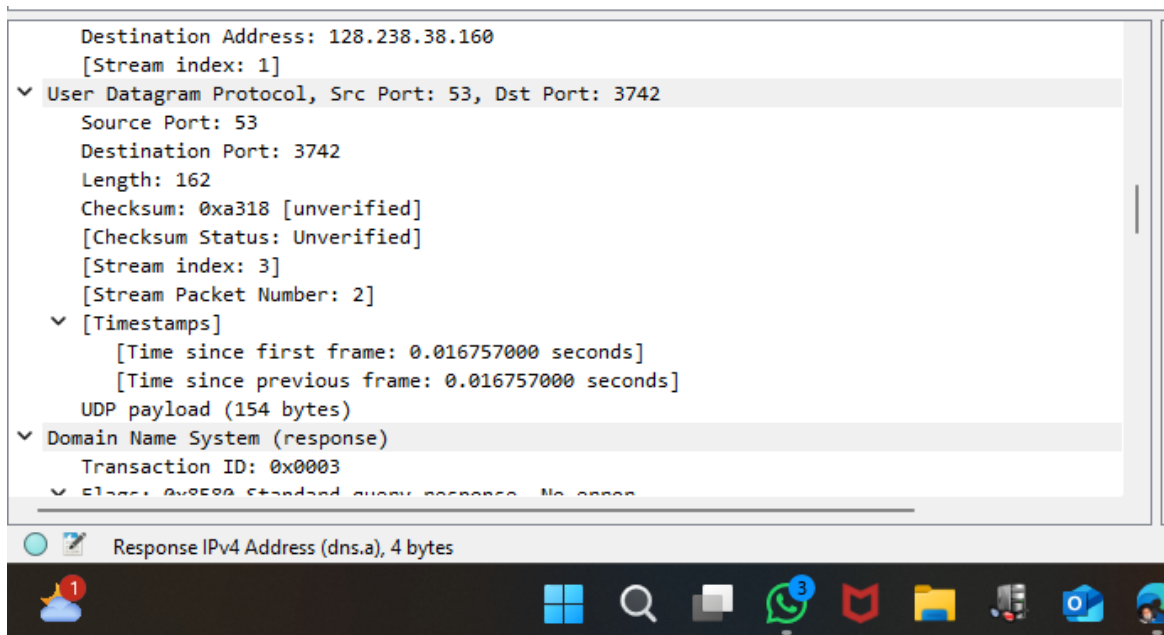
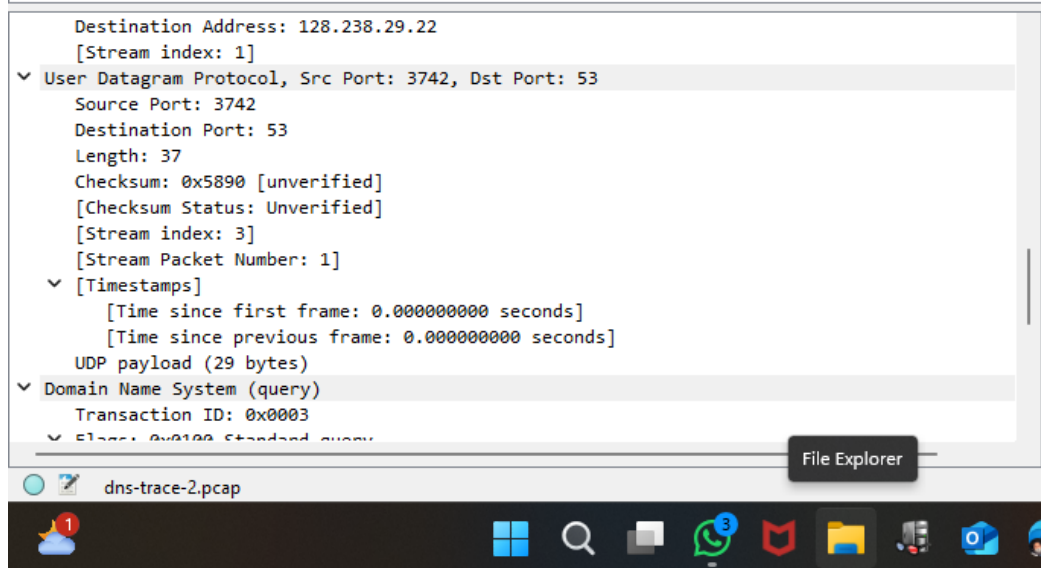


7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?
No, the host does not issue new DNS queries for each image. The DNS information is cached after the initial query for the domain."

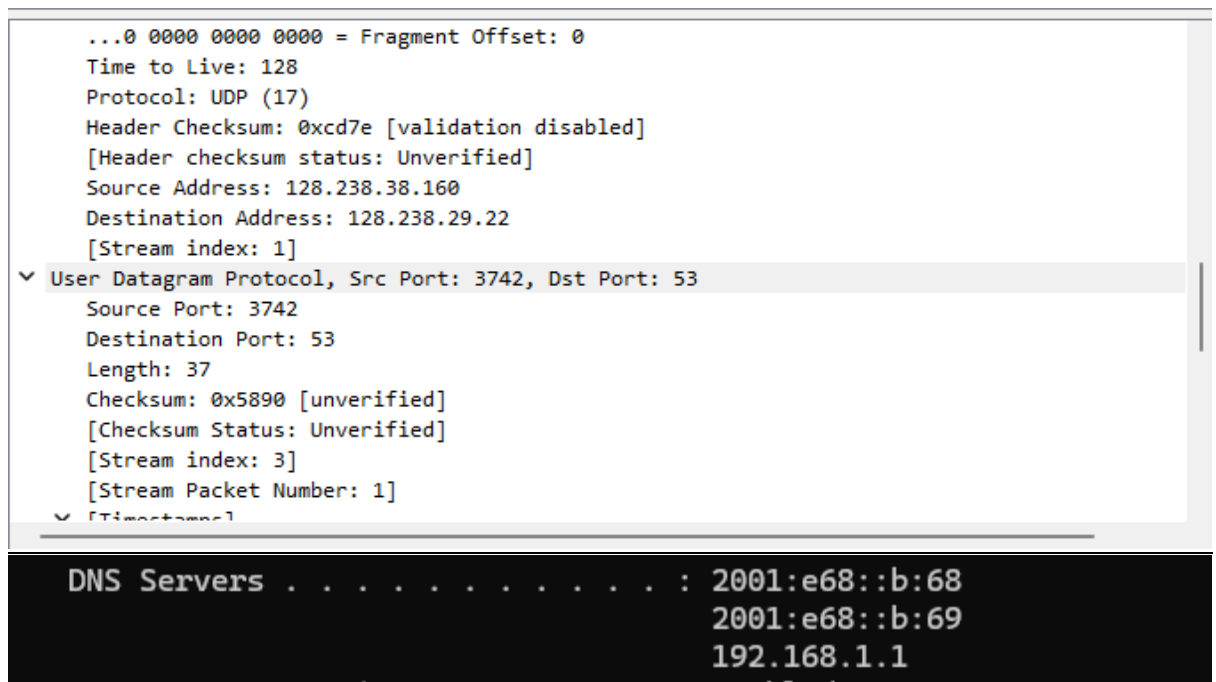
- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup sent three DNS queries and received three DNS responses. For this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

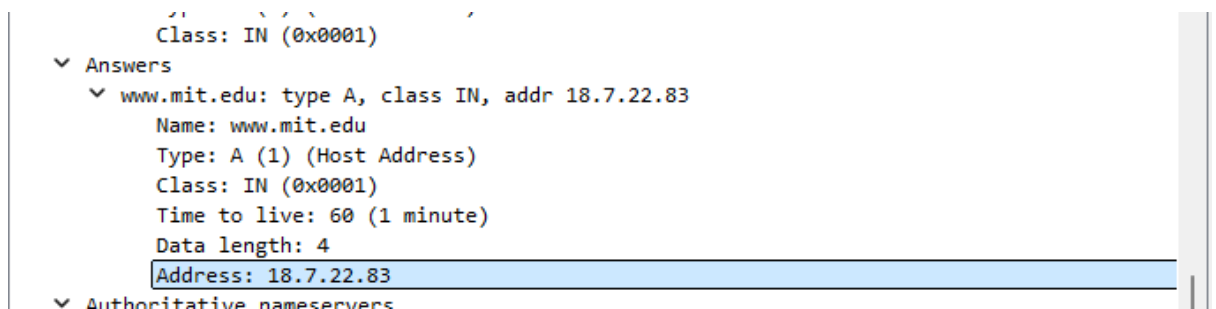
Destination port DNS query message: 53, Source port DNS response message: 53



9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.
128.238.29.22. No, if the destination IP address differs



10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.
Type “A”. Yes.



11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.
Only 1 answer provided.

```
Class: IN (0x0001)
  Answers
    www.mit.edu: type A, class IN, addr 18.7.22.83
      Name: www.mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 60 (1 minute)
      Data length: 4
      Address: 18.7.22.83
  Authoritative nameservers
```